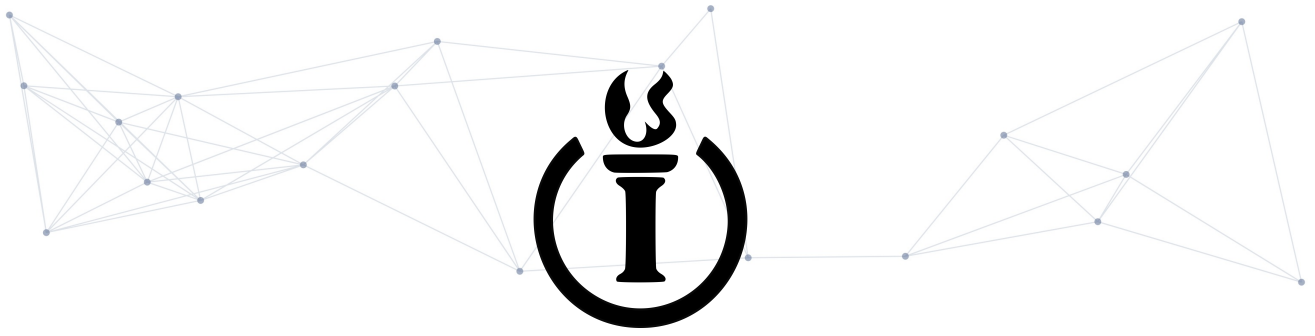




International Computer Science Competition

Problem Set for Ambassadors

12 May 2026

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For your ambassador application, please submit solutions to the following problems:

Problem: Hashing Basics

What is the purpose of a hash function in computer science? Why is it important for a hash table to have a good hash function?

Problem: Boolean Logic

You have three switches: A, B, and C. The light turns on only when exactly one of them is on. Write a Boolean expression for this condition.

Problem: Count Even Numbers

Describe an algorithm that counts how many even numbers are in a list. You can describe the algorithm in human language, pseudo code, or your favorite programming language.

